LESSON PLAN

ROCKET ENGINE (14)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	01
Total No. of Lesson Plans	2	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Introduction to non-breathing engine, solid rocket engines its advantages and disadvantages
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Introduction to non-breathing engine, solid rocket engines its advantages and disadvantages
5. Specific Objectives	14.1.1.1 Trainee will be able to explain types of rocket engine and their working principle.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of Solid Rocket Motor Diagrams; Safety: Do not handle propellant samples; Instructor supervision required for diagram analysis
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a rocket engine?
(b) Define a non-breathing engine in one sentence.
(c) List two common types of rocket engines.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of non-breathing engine • Working principle of solid rocket engines • Components of solid propellant grain • Ignition and combustion process • Thrust generation mechanism	L & Ds	PPT, IB	15 min
(b)	• Advantages of solid rocket engines • Disadvantages of solid rocket engines • Comparison with liquid engines • Storage and handling characteristics • Reliability and simplicity factors	L & Ds	PPT, PDF	20 min
(c)	• Real-world applications of SRMs • Limitations in thrust control • Safety protocols for solid propellants • Environmental considerations • Future trends in solid propulsion	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain why solid rocket engines are considered non-breathing.
(b) Identify one major advantage and one major disadvantage of solid rocket engines.
(c) Describe the working principle of a solid rocket engine using the grain structure.

12. Home Assignment and Follow Up	Prepare a brief note comparing the thrust control capabilities of solid rocket engines versus liquid rocket engines to be discussed in Lesson Plan 02.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Introduction to liquid rocket engines its advantages and disadvantages.	Refer to the master syllabus reference list	LP 02

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min